PHYSICS

1) Two trains, each of length 200 m are running on parallel track. One train overtakes the other in 20 s and one crosses the other in 10 s. The their velocities are:-

(1) 30 ms^{-1}, 20 ms^{-1}
(2) 20 ms^{-1}, 10 ms^{-1}
(3) 5 ms^{-1}, 15 ms^{-1}
(4) 30 ms^{-1}, 10 ms^{-1}

2) A cyclist starts from rest and moves with a constant acceleration of 1 m/s^2. A boy who is 48 m behind the cyclist starts moving with a constant velocity of 10 m/s. After how much time cyclist overtakes boy ?

(1) 8s
(2) 12s
(3) 10s
(4) both (1) and (2)

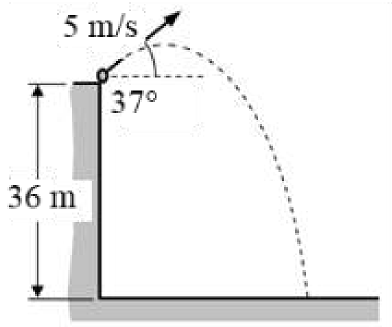
3) A projectile has a time of flight T and range R. If time of flight is doubled keeping angle of projection same, then what will be the new range :-

(1) R
(2) 2R
(3) R/2
(4) 4R

4) The horizontal range of projectile is four times the maximum height. The value of angle of projection is :-

(1) 90°
(2) 60°
(3) 45°
(4) 30°

5) A ball is thrown from the top of 36 m high tower with velocity 5 m/s at an angle of 37° above the horizontal as shown. Its horizontal range on the ground is closest to [take g = 10 m/s^2]



(1) 12 m

(2) 18 m

(3) 24 m

(4) 30 m

6) A man is walking on a road with a velocity of 3km/h when suddenly, it starts raining with velocity 10 km/h in vertically downward direction, relative velocity of the rain with respect to man is :

(1) $\sqrt{13}$ km/hr

(2) $\sqrt{7}$ km/hr

(3) $\sqrt{109}$ km/hr

(4) 13 km/h

7) Rain is falling vertically downwards with a speed of 4 km/h. A girl moves on a straight road with a velocity of 3 km/h. The apparent velocity of rain with respect to the girl is :-

(1) 3 km/h

(2) 4 km/h

(3) 5 km/h

(4) 7 km/h

8) A man is crossing a river flowing with velocity of 5 m/s. He reaches a point directly across at a distance of 60 m in 5 sec. His velocity in still water should be -

(1) 12 m/s

(2) 13 m/s

(3) 5 m/s

(4) 10 m/s

9) A swimmer wishes to cross a 500 m wide river flowing at 5 km/hr. His speed with respect to river is 3 km/hr. The minimum time to cross the river is :-

(1) 20 min

(2) 6 min

(3) 7.5 min

(4) 10 min

10) A person climbs up a stalled escalator in 60 s. If standing on the same but escalator running with constant velocity he takes 40 s. How much time is taken by the person to walk up the moving escalator?

(1) 45 s
(2) 27 s
(3) 37 s
(4) 24 s

11) Two cars A and B moving with speed 50KM/H and 100 KM/Hr respectively on two different roads having angle 60° between them. Find the speed of car B as observed by driver of car A.

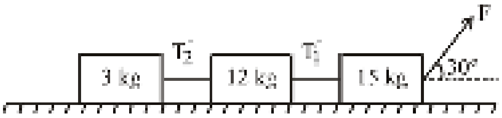
(1) 50 KM/Hr
(2) $50\sqrt{2}$ KM/Hr
(3) $50\sqrt{3}$ KM/Hr
(4) $\frac{200}{\sqrt{3}}$ KM/Hr

12) A boat moves relative to still water with a velocity which is n times the river flow velocity. At what angle to the stream direction must be boat move to cross the river along shortest path ?

(1) $\frac{\pi}{2}$
(2) $\sin^{-1}\left(\frac{1}{n}\right)$
(3) $\frac{\pi}{2} + \sin^{-1}\left(\frac{1}{n}\right)$
(4) $\frac{\pi}{2} - \sin^{-1}\left(\frac{1}{n}\right)$

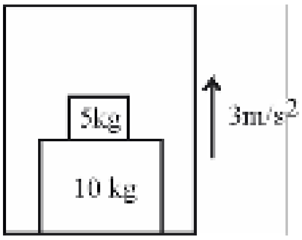
13) A solid sphere of mass 10 kg is placed over two smooth inclined planes as shown in figure. Normal reaction at 1 and 2 will be : ($g = 10\ m/s^2$)

(1) $50\sqrt{3}N$, 50 N
(2) 50 N, 50 N
(3) 50 N, $50\sqrt{3}N$
(4) 60 N, 40 N

14) The surface is frictionless then the ratio of T_1 and T_2 is : 

(1) $\sqrt{3} : 1$
(2) $1 : \sqrt{3}$
(3) 1 : 5
(4) 5 : 1

15) The force exerted on 10kg block by floor of lift, as shown in the figure is (take g = $10m/s^2$) :-

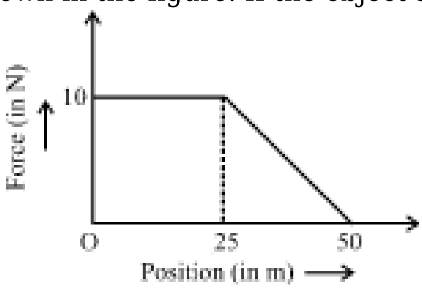


(1) 180N
(2) 150N
(3) 195N
(4) 135N

16) If an inclined plane is made slowly horizontal by reducing its inclination with horizontal, the component of weight of block resting on the inclined plane parallel to the plane :

(1) Remains same
(2) Increases
(3) Deceases
(4) First increases and then decreases

17) An object of mass 5kg is acted upon by a force that varies with the position of the object as shown in the figure. If the object starts out from rest at the point x = 0, then its speed at x = 25m is



:-

(1) 5 ms^{-1}
(2) 10 ms^{-1}
(3) 20 ms^{-1}
(4) 28 ms^{-1}

18) A particle of mass 2kg moves in free space such that its position vector varies with time as

$\vec{r} = \left[\left(3+4t^2\right)\hat{i} + (2t)\hat{j} + (3-6t)\hat{k}\right]$ m where t is in seconds. Net force acting on the particle is

(1) zero
(2) parallel to x-axis
(3) parallel to y-axis
(4) time dependent

19) A man of mass 60 kg is riding in a lift. The weight of the man recorded by a weighing machine, when the lift is accelerating upwards and downwards at 2 ms^{-2}, are respectively (take, g = 10 ms^{-2}) :-

(1) 720 N and 480 N
(2) 480 N and 720 N
(3) 600 N and 600 N
(4) none to these

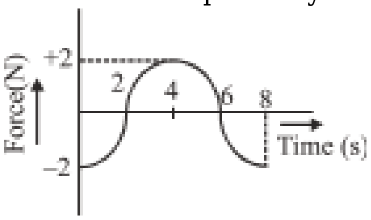
20) A hammer of mass 1 kg strikes a wall and comes to rest in 0.1s. If the retarding force that stops the hammer is 60 N, speed of the hammer was :

(1) 3 m/s
(2) 6 m/s
(3) 300 m/s
(4) 600 m/s

21) A train has 10 wagons each of mass 1000 kg attached to it. They are being pulled by force 10^4 N. Find out force exerted on last four wagons.

(1) 4×10^3 N
(2) 4×10^4 N
(3) 5×10^4 N
(4) 5×10^3 N

22) The force-time(F–t) curve of a particle executing linear motion is as shown in the figure. The momentum acquired by the particle in time interval from zero to 8 second will be :-



(1) –2 N-s
(2) + 4 N-s
(3) 6 N-s
(4) Zero

23) A particle of mass 2kg is moving on a smooth horizontal surface with a velocity of 10 m/s. If a force of 20 N is applied on it then velocity of particle after 5 second is :

(1) 50 m/s
(2) 40 m/s
(3) 60 m/s
(4) 10 m/s

24) The linear momentum p of a body moving in one dimension varies with time according to the equation $p = a + bt^2$, where a and b are positive constants. The net force acting on the body is :-

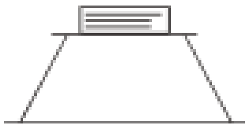
(1) Proportional to t^2
(2) A constant
(3) Proportional to t
(4) Inversely proportional to t

25) Newton's II law of motion connects :-

(1) momentum and acceleration
(2) change of momentum and velocity
(3) rate of change of momentum and external force
(4) rate of change of force and momentum

26) If you push a book by applying a force $\vec{F} = 3\hat{i} + 4\hat{j}$, the force which is exerted by the book on you is :-

(1) $3\hat{i} + 4\hat{j}$
(2) $3\hat{i} - 4\hat{j}$
(3) $-3\hat{i} - 4\hat{j}$
(4) $\hat{i} + \hat{j}$

27) A book is at rest on a table. What is the "reaction" force according to Newton's third law to the gravitational force by the earth on the book ? 

(1) The normal force exerted by the table on the book
(2) The normal force exerted by the table on the ground
(3) The normal force exerted by the ground on the table
(4) The gravitational force exerted on the earth by the book

28) **Assertion (A) :** Every mutual interaction between two bodies always follow the action–reaction law.
Reason (R) : Single isolated force without reaction is not possible

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

29) Match the following:

	List-I		List-II
(a)	Newton's third law	(i)	Gives the magnitude of force
(b)	Newton's first law	(ii)	Law of conservation of momentum
(c)	Newton's second law	(iii)	Defines inertia and gives the concept of force

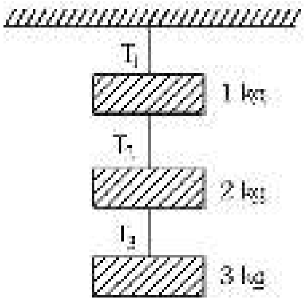
The correct option is

(1) a → ii; b → iii; c → i

(2) a → iii; b → ii; c → i

(3) a → i; b → ii; c → iii

(4) a → ii; b → i; c → iii



30) Find the tension T_2 in the system shown in fig.

(1) 1g N

(2) 2g N

(3) 5g N

(4) 6g N

31) **Statement-1 :** Weight and normal reaction on a block when placed over a horizontal surface form action-reaction pair.
Statement-2 : Action and reaction pair must act on the same object.

(1) Statement-1 is true, Statement-2 is true

(2) Statement-1 is false, Statement-2 is true

(3) Statement-1 is true, Statement-2 is false

(4) Statement-1 is false, Statement-2 is false

32) **Statement-1:** If heavy body and light body have same momentum. Then they also have same K.E.
Statement-2: Kinetic energy is independent of mass of the body.

(1) Both statement 1 and statement 2 are true and the statement 2 is the correct explanation of the statement 1.

(2) Both statement 1 and statement 2 are true but statement 2 is not the correct explanation of the statement 1.

(3) The statement 1 and statement 2 both are false.

(4) Statement 1 is true but statement 2 is false.

33) A ball is thrown vertically upwards with a certain velocity. Neglecting air resistance, identify the wrong statement(s) regarding the motion of the ball.
(A) During the upward motion net force on the ball acts upwards.
(B) During the downward motion net force on the ball acts downwards.
(C) At the highest position no force acts on the ball.
(D) The net force on the ball always act in the downward direction.

(1) A, B

(2) B, C

(3) B, D

(4) A, C

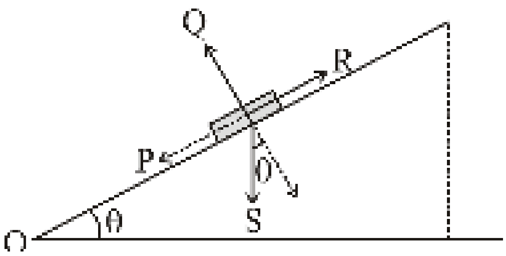
34) **Assertion :-** A table cloth can be pulled from a table without dislodging the dishes.
Reason :- To every action there is an equal and opposite reaction.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

35) 

When a body of mass 'm' just at the verge of sliding as shown, match list-I with list-II.

	List-I		List-II
(a)	Normal reaction	(i)	P
(b)	Frictional force (f_s)	(ii)	Q
(c)	Weight (mg)	(iii)	R
(d)	mgsinθ	(iv)	S

Choose the correct answer from the options given below:

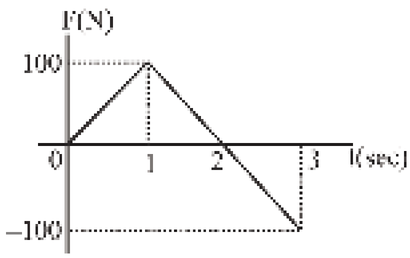
(1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)

(2) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(3) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(4) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

36) For the given force-time graph, impulse acting in first 3 second is :-



(1) 50 N-sec

(2) 150 N-sec

(3) 100 N-sec

(4) 200 N-sec

37) A person is sitting in a travelling train and facing engine. He tosses up the coin and coin falls behind him. It can be concluded that train is :-

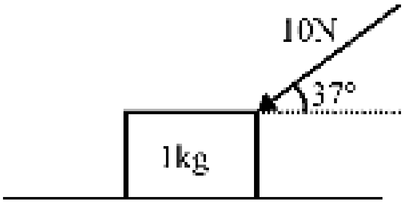
(1) Moving forward and gaining speed

(2) Moving forward and losing speed

(3) Moving forward with uniform speed

(4) Moving backward with uniform speed

38) A block is kept on a smooth horizontal surface and a force is applied on it as shown in figure. The acceleration of the block will be ($\sin 37° = 3/5$ and $g=10\ m/s^2$)



(1) 10 m/s^2 in the direction of 10 N force

(2) 10 m/s^2 in horizontal direction

(3) 8 m/s^2 in horizontal direction

(4) 6 m/s^2 in vertical direction

39) A body of mass 6 kg moves in a straight line according to the equation $x = t^3 – 75t$
where x denotes the distance in metre and t the time in second. The force on the body at t = 4 second is :–

(1) 64 newton

(2) 72 newton

(3) 144 newton

(4) 36 newton

40) At an instant, mass of rocket is 500 kg. If it is moving with constant velocity of 1 km/s, then find the rate of fuel consumption.

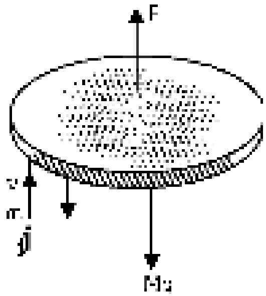
(1) 10 m/s

(2) 5 kg/sec

(3) 1 kg/sec

(4) 50 kg/sec

41) A disc of mass 1·0 kg kept floating horizontally in air by firing bullets of mass 0·05 kg each vertically at it, at the rate of 10 per second. If the bullets rebound with the same speed, the speed with which these are fired will be–



(1) 0·098 m/s

(2) 0·98 m/s

(3) 9·8 m/s

(4) 98·0 m/s

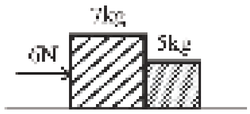
42) A stone is accelerated upwards by a cord whose breaking strength is three times the weight of the stone. The maximum acceleration with which the stone can be moved up without breaking the cord is :

(1) g

(2) 2g

(3) 3g

(4) 4g

43) Two blocks of masses 7 kg and 5 kg are placed in contact with each other on a smooth surface. If a force of 6 N is applied on the heavier mass, the force on the lighter mass is :- 

(1) 3.5 N

(2) 2.5 N

(3) 7 N

(4) 5 N

44) A rocket consumes fuel at the rate of 100 kg/s. The exhust gases are ejected at a speed of 5 × 10^4 m/s. Neglecting the effect of gravity, the force experienced by the rocket is :-

(1) 5 × 10^2 N

(2) 5 × 10^4 N

(3) 5 × 10^6 N

(4) 50 N

45) A body of mass 5 kg is suspended by string making angle 60° & 30° with horizontal then –

(a) T_1 = 25 N

(b) T_2 = 25 N

(c) $T_1 = 25\sqrt{3}$ N

(d) $T_2 = 25\sqrt{3}$ N

(1) a, b

(2) a, d

(3) c, d

(4) b, c

CHEMISTRY

1)

$C(s) + O_2(g) \rightleftharpoons CO_2(g)$,

The given reaction is started with 1 mol C, 4 mol O_2 and 3 mol CO_2. If $K_p = \frac{1}{5}$ then moles of CO_2 at equilibrium will be :-

(1) 2.31

(2) 1.17

(3) 5.28

(4) 8.29

2) The equilibrium :
$P_4(g) + 6Cl_2(g) \rightleftharpoons 4PCl_3(g)$
is attained by mixing equal moles of P_4 and Cl_2 in an evacuated vessel. Then at equilibrium which of the following is always correct :-

(1) $[Cl_2] > [PCl_3]$

(2) $[Cl_2] > [P_4]$

(3) $[P_4] > [Cl_2]$

(4) $[PCl_3] > [P_4]$

3) At a certain temperature, only 50% HI is dissociated into H_2 and I_2 at equilibrium. The equilibrium constant for $2HI_{(g)} \rightleftharpoons H_2(g) + I_2(g)$ is :-

(1) 1.0

(2) 3.0

(3) 0.5

(4) 0.25

4) In the following reversible reaction

$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$
more suitable condition for the higher production of SO_3 is :-

(1) High pressure
(2) Low pressure
(3) Removal of SO_2
(4) Removal of O_2

5) For the reaction $2A(g) \rightleftharpoons B(g) + 3C(g)$, at a given temperature, $K_c = 16$. What must be the volume of the flask, if a mixture of 2 mole each of A, B and C exist in equilibrium ?

(1) $\frac{1}{4}L$
(2) $\frac{1}{2}L$
(3) 1L
(4) None of these

6) When 3 mole of A and 1 mole of B are mixed in 1 litre vessel the following reaction take place $A(g) + B(g) \rightleftharpoons 2C(g)$. 1.5 mole of C are formed. The equilibrium constant for the reaction is :-

(1) 0.12
(2) 0.25
(3) 0.50
(4) 4.0

7) The value of $\frac{K}{Q}$ is 0.33. The reaction will proceeds in ______direction.

(1) Forward
(2) Backward
(3) Not any
(4) None of these

8) In a chemical reaction, equilibrium is established when :-

(1) Opposing reaction stops
(2) Concentrations of reactants and products become equal
(3) Rate of backward reaction is same as that of rate of forward reaction ($r_f = r_b$)
(4) Reaction stops

9) $A_{(s)} \rightleftharpoons B_{(g)} + C_{(g)}$
In the given reaction, at equilibrium, total pressure of the system is 0.6 atm. Then find out K_p.

(1) 0.3
(2) 0.09

(3) 3

(4) 0.6

10) Consider the following gaseous equilibria with equilibrium constant K_1 and K_2 respectively.

$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

The equilibrium constant is related as :-

(1) $2K_1 = K_2^2$

(2) $K_1^2 = \frac{1}{K_2}$

(3) $K_2^2 = \frac{1}{K_1}$

(4) $K_2 = \frac{2}{K_1^2}$

11) For a chemical reaction of the type $A \rightleftharpoons B$, K = 2.0 and $B \rightleftharpoons C$, K = 0.01. Equilibrium constant for the reaction $2C \rightleftharpoons 2A$ is :-

(1) 25

(2) 50

(3) 2.5×10^3

(4) 4×10^{-4}

12) On increasing temperature, the equilibrium constant of exothermic and endothermic reactions respectively:-

(1) Increases and Decreases

(2) Decreases and Increases

(3) Increases and Increases

(4) Decreases and Decreases

13)

Choose the incorrect radii order :-

(1) $Ti^{2+} < Ti^{3+} < Ti^{4+}$

(2) $S^{2-} > S^{-}$

(3) $Mg^{2+} < C^{4-}$

(4) $H^{+} < H^{-}$

14) An element having atomic number 120 have not yet been discovered. What will be its position in long form of periodic table ?

(1) 2^{nd} group and 7^{th} period

(2) 2^{nd} group and 8^{th} period

(3) 3^{rd} group and 6^{th} period

(4) 3^{rd} group and 7^{th} period

15) Correct order of 2^{nd} I.P. of C, N, O, F would be -

(1) C < N < F < O

(2) C < O < N < F

(3) N < C < F < O

(4) N < C < O < F

16) **Assertion (A) :-** Elements of a group exhibits similar chemical properties.
Reason (R) :- Elements of a group possess similar electronic configuration of their outermost shell.

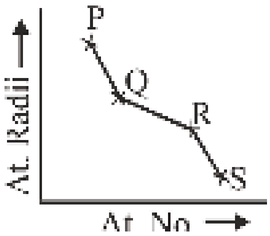
(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

17) In graph P, Q, R, S are element of same period then element R may be :-



(1) B

(2) C

(3) N

(4) O

18) In Mendeleev periodic table, correction were done in atomic weight of following element :-

(1) Be

(2) In

(3) Pt

(4) All of the above

19) Which of the following configuration has minimum I^{st} ionization potential :

(1) $1s^2\ 2s^1$

(2) $1s^2\ 2s^2\ 2p^2$

(3) $_{36}[Kr]\ 4d^{10}\ 5s^2\ 5p^5$

(4) $_{54}[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^5$

20) Correct order of first ionisation potential of C, N, O, F :-

(1) C > N > O > F

(2) C < N < O < F

(3) F > N > O > C

(4) N > O > F > C

21) Consider the isoelectronic species, Na^+, Mg^{2+}, F^- and O^{2-}. The correct order of increasing order of their radii is.

(1) $F^- < O^{2-} < Mg^{2+} < Na^+$

(2) $Mg^{2+} < Na^+ < F^- < O^{2-}$

(3) $O^{2-} < F^- < Na^+ < Mg^{2+}$

(4) $O^{2-} < F^- < Mg^{2+} < Na^+$

22) Atomic number 34 belongs to :-

(1) 3^{rd} period, 15 group, p block

(2) 4^{th} period, 16 group, p block

(3) 4^{th} period, 2 group, s block

(4) 5^{th} period, 12 group, d block

23) After the filling of np orbital next orbital filled will be :

(1) ns

(2) np

(3) (n–1)d

(4) (n+1)s

24) X = $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
Find out the group of element X.

(1) 2^{nd}

(2) 8^{th}

(3) 7^{th}

(4) 4^{th}

25) The size of the species, Pb, Pb^{2+}, Pb^{4+} decreases as :-

(1) $Pb^{4+} > Pb^{2+} > Pb$

(2) $Pb > Pb^{2+} > Pb^{4+}$

(3) $Pb > Pb^{4+} > Pb^{2+}$

(4) $Pb^{4+} > Pb > Pb^{2+}$

26) Which of the following have nearly same atomic radius :-

(1) Na, K, Rb, Cs

(2) Li, Be, B, C
(3) Fe, Co, Ni
(4) F, Cl, Br, I

27) If total 100 elements are present in periodic table then how many of them contain e^- in 4d subshell.

(1) 60
(2) 62
(3) 64
(4) 80

28) Match the following :-

(1)	$(n-2)f^{14}(n-1)d^4\ ns^2$	(a)	Noble gas
(2)	Ga	(b)	Halogen
(3)	Ar	(c)	Transition element
(4)	Cl	(d)	Eka aluminium

(1) 1-c, 2-d, 3-a, 4-b
(2) 1-a, 2-b, 3-c, 4-d
(3) 1-c, 2-d, 3-b, 4-a
(4) 1-b, 2-a, 3-d, 4-c

29) Which of the following has highest Z_{eff} ?

(1) H^-
(2) Li^+
(3) Be^{+2}
(4) B^{+3}

30) Which of the following radius order is not due to lanthanoid contraction?

(1) $Ag \cong Au$
(2) $A\ell \simeq Ga$
(3) $Ru \cong Os$
(4) $Zr \simeq Hf$

31) Position of noble gases in long form of periodic table according to IUPAC :-

(1) Zero group
(2) 18th group
(3) VIII A
(4) Absent in long form of periodic table

32) Size of Mo is very similar to W due to -

(1) Difference of atomic number by one
(2) Contraction in size in first transition series element.
(3) Lanthanide contraction
(4) actinide contraction

33) Select incorrect statement about element Uut :-

(1) It belongs to p-block.
(2) Atomic number of this element is 113.
(3) IUPAC name of the element is ununtrium.
(4) It belongs to d-block.

34) The decreasing order of Z_{eff} will be :-

(1) $O^{-2} < O < O^{+} < O^{+2}$
(2) $O^{-2} > O > O^{+} > O^{+2}$
(3) $O^{+2} > O^{+} > O > O^{-2}$
(4) $O < O^{-2} < O^{+} < O^{+2}$

35) Maximum number of unpaired electrons are in :-

(1) Na^{+}
(2) F
(3) O^{-2}
(4) N

36) IP_1 and IP_2 of Mg are 178 and 348 k.cal mol^{-1}. The enthalpy required for the reaction
$Mg_{(g)} \rightarrow Mg^{+2}_{(g)} + 2e^{-}$

(1) + 170 k.cal
(2) + 526 k.cal
(3) - 170 k.cal
(4) -526 k.cal

37) (a) The long form periodic table is based on increasing atomic weight.
(b) First d-block series starts with element scandium and ends with zinc.
(c) Alkali and alkaline earth metals are s-block elements.
(d) Chalcogens and halogens are p-block elements.
The correct statements are :-

(1) a, b and c
(2) a and b
(3) b, c and d
(4) all

38) Electronic configuration of X^{2+} and Y^{3+} are : X^{2+} = $[Ar]3d^8$,Y^{3+}=$[Ar]3d^3$. What are the atomic number of X and Y respectively ?

(1) 28, 24
(2) 28, 25
(3) 28, 26
(4) 28, 27

39) The covalent and vander waal's radii of hydrogen respectively are :

(1) 0.37 Å, 0.8 Å
(2) 0.37 Å, 0.37 Å
(3) 0.8 Å, 0.8 Å
(4) 0.8 Å, 0.37 Å

40) If the Z_{eff} of Na is x then Ca will be :-

(1) x
(2) x + 0.35
(3) x + 0.65
(4) x + 1.05

41) Match the following :-

	Outer configuration		**IP values (ev/atom)**
a	ns^1	P	19, 192,210, 230
b	ns^2	Q	24, 36, 42, 250, 270
c	ns^2np^1	R	22, 32, 200, 240
d	ns^2np^2	S	28, 34, 48, 54, 280

(1) a-S, b-R, c-Q, d-P
(2) a-P, b-R, c-Q, d-S
(3) a-Q, b-R, c-S, d-P
(4) a-R, b-P, c-S, d-Q

42)

Correct order of atomic radii in group 13 elements is :-

(1) B < Al < In < Ga < Tl
(2) B <Al < Ga < In < Tl
(3) B < Ga < Al < TI < In
(4) B< Ga < Al < In <TI

43) The diagram below is a part of the skelton of the periodic tables in which elements are indicated by letter which are not their usual symbols.

	D	
A	C	F
52	B	E

The correct option is :-

(1) F is krypton

(2) 'B' exist in nature in liquid state

(3) 'A' is a 5th period element in a periodic table

(4) 'E' exist as diatomic gas

44) Match the columns :-

Column-I (Atomic No.)		Column-II (Block)	
(a)	15	(P)	s-block
(b)	19	(q)	p-block
(c)	25	(r)	d-block
(d)	64	(s)	f-block

(1) a–(q), b–(p), c–(r), d–(s)

(2) a–(r), b–(p), c–(q), d–(s)

(3) a–(p), b–(r), c–(q), d–(s)

(4) a–(q), b–(r), c–(p), d–(s)

45) If the atomic number of an element is 58, it will be placed in the periodic table in the :-

(1) III B group and 6th period

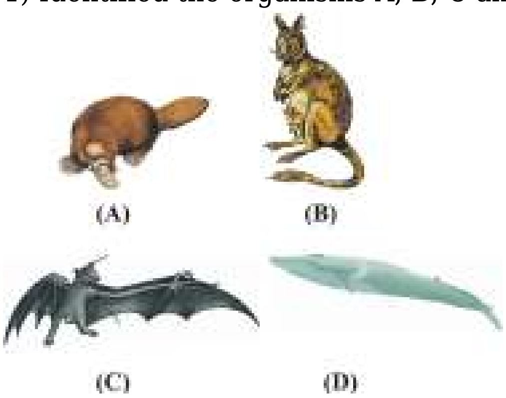
(2) IV B group and 6th period

(3) V B group and 7th period

(4) none of these

BIOLOGY

1) Identified the organisms A, B, C and D.



(1) Balaenoptera, Macropus, Pteropus & Ornithorhynchus

(2) Ornithorhynchus, Macropus, Pteropus & Balaenoptera

(3) Ornithorhynchus, Macropus, Balaenoptera & Pteropus

(4) Ornithorhynchus, Balaenoptera, Pteropus & Macropus

2) List the phyla in the correct order of their placement in classification
I. Chordata II. Annelida
III. Arthropoda IV. Platyhelminthes
V. Ctenophora VI. Aschelminthes

(1) VI, I, V, IV, III, II

(2) II, III, IV, V, VI, I

(3) V, IV, VI, II, III, I

(4) III, II, VI, I, V, IV

3) Which of the following statements (a - e) are correct ?
a. the pelvic fins of female shark bear claspers
b. in *Obelia*, polyps produce medusae sexually and medusae form the polyps asexually
c. In birds hind limbs are modified into wings
d. In reptiles body is covered with dry and cornified skin
e. beak is found in birds

(1) b, d, e

(2) a, c, e

(3) d, e

(4) a, b, c

4) Identify the correct match from Column - I and Column - II :

Column - I		**Column - II**	
(a)	*Scoliodon*	(i)	Saw fish
(b)	*Pristis*	(ii)	Sting ray
(c)	*Trygon*	(iii)	Electric ray
(d)	*Torpedo*	(iv)	Dog fish

Options:

(1) a-i, b-ii, c-iii, d-iv

(2) a-ii, b-iii, c-iv, d-i

(3) a-iv, b-iii, c-i, d-ii

(4) a-iv, b-i, c-ii, d-iii

5) The structure given below is present in which animal ?

(1) *Scypha*
(2) *Ctenoplana*
(3) *Physalia*
(4) *Taenia*

6) A student brought home a strange animal which he found outside under a rock. It had moist skin, a complete digestive tract, a ventral nerve cord, and have metameric segments. Identify the phylum of the animal.

(1) Porifera
(2) Annelida
(3) Mollusca
(4) Echinodermata

7) Which of the following is incorrectly matched :-

(1) Hydra, Gorgonia, Jelly fish ⇒ Radial symmetry
(2) Pleurobrachia, Ctenoplana, Sycon ⇒ Radial symmetry
(3) Earthworm, Fishes, Scorpion ⇒ Bilateral symmetry
(4) Sycon, Spongilla, Euspongia ⇒ Asymmetry

8) In water canal system of porifera, water flows through which one of the following ways :-

(1) Ostia → Spongocoel → Osculum → Exterior
(2) Spongocoel → Ostia → Osculum → Exterior
(3) Osculum → Spongocoel → Ostia → Exterior
(4) Osculum → Ostia → Spongocoel → Exterior

9) **Statement-I :-** All phyla from platyhelminthes to chordata are triploblastic.
Statement-II :- All phyla from annelida to chordata are true coelomate.

(1) Statement I is correct but II is incorrect.
(2) Statement II is correct but I is incorrect.
(3) Both statements are correct.
(4) Both statements are incorrect.

10) Read the assertion and reason carefully to mark the correct option out of the options given

below :
Assertion : A closed circulatory system is found in annelids.
Reason : Annelids possess true coelom.

(1) Both the assertion and the reason are true and the reason is a correct explanation of the assertion.

(2) Both the assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both the assertion and reason are false.

11) Ctenophores have -
(a) Exclusively marine
(b) Exhibit bioluminescence
(c) Body bears eight internal rows of ciliated comb plates.
(d) Internal fertilization.

(1) (a), (b) and (c)

(2) (a) and (b) only

(3) (b), (c) and (d)

(4) All of the above

12) Which is not true for reptiles?

(1) They respire by lungs

(2) All have 4 – chambered heart

(3) They are chordate animals

(4) They have closed type of circulatory system

13) Which is not a sponges?

(1) Sycon

(2) Spongilla

(3) Scypha

(4) Sea anemone

14) Coxal glands are the excretory structures of :

(1) Cockroach

(2) Scorpion

(3) Prawn

(4) Leech

15) Chitinuous exoskeleton is found in which phylum?

(1) Annelida

(2) Porifera

(3) Arthropoda
(4) Echinodermata

16) Out of following in which phylum only sexual reproduction occurs ?

(1) Protozoa
(2) Porifera
(3) Coelenterata
(4) Ctenophora

17) Metamerism is first observed in :-

(1) Platyhelminthes
(2) Aschelminthes
(3) Annelida
(4) Arthropoda

18) Closed type of circulatory system is present in

(1) Annelida and arthropod
(2) Arthropod and mollusc
(3) Annelida and chordates
(4) Mollusc and echinodermates

19) In which animal soft body surrounded by an external calcareous shell?

(1) Annelid
(2) Mollusc
(3) Echinoderm
(4) Poriferan

20) Animals which exhibits alternation of generation :-

(1) Hydra
(2) Obelia
(3) Adamsia
(4) All of these

21) Which one is not correct matched pairs from following?

(1) Echinus → Sea urchin → Echinodermata
(2) Pinctada → Pearl oyster → Mollusca
(3) Balanoglosus → Lancelet → Cephalochordata
(4) Loligo → Squid → Mollusca

22) Organ level of organisation first occurred in?

(1) Aschelminthes
(2) Platyhelminthes
(3) Annelida
(4) Arthropoda

23) "Animals exhibits radial or bilateral symmetry depending on the stage". This statement is:-

(1) True for Echinodermates
(2) True for Molluscans
(3) True for Arthropods
(4) True for Hemichordates

24) The name of cells, which line the spongocoel?

(1) Archaeocytes
(2) Scleroblast
(3) Choanocytes
(4) Pinacocytes

25) **Assertion (A) :** The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus etc. form a complex structure called the limbic lobe or limbic system.
Reason (R) : Limbic lobe responsible for complex functions like intersensory associations, memory & communications.
Choose the correct choice :-

(1) Both **(A)** and **(R)** are correct, and **(R)** is correct explanation of **(A)**
(2) Both **(A)** and **(R)** are correct, but **(R)** is not correct explanation of **(A)**
(3) **(A)** is correct while **(R)** is not correct.
(4) Both **(A)** & **(R)** are incorrect.

26) **Statement I:-** In haemodialysis, blood drained from a convenient vein is pumped into a dialysing unit after adding anti-heparin.
Statement II:- The dialysis unit contains a coiled cellophane tube surrounded by dialysis fluid.

(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

27) Which of the following would not be a part of glomerular filtrate in a normal heathy person

(1) RBC
(2) Plasma protein
(3) Glucose
(4) Both (1) & (2)

28) Secretion of ANF is stimulated by :
(ANF = Atrial Natriuretic Factor)

(1) Increased Blood sugar level
(2) Decreased blood pressure
(3) Low Ca^{+2} level in Blood
(4) Increased Blood pressure

29)

Match the columns

Column-I		**Column-II**	
A	Sweat glands	I	Sterols , hydrocarbons
B	Sebaceous glands	II	Carbondioxide
C	Liver	III	Cholesterol , bilirubin
D	Lungs	IV	NaCl , small amount of urea

	A	**B**	**C**	**D**
(1)	I	III	IV	II
(2)	III	I	IV	II
(3)	IV	I	III	II
(4)	IV	III	II	I

(1) 1
(2) 2
(3) 3
(4) 4

30)

Select the **correct** statements.
(a) Angiotensin II activates the adrenal gland to release aldosterone.
(b) Aldosterone leads to increase in blood pressure.
(c) ANF acts as a check on renin-angiotensin mechanism.
(d) ADH causes vasodilation.
(e) ADH is released from kidney.
Choose **the most appropriate answer** from the options given below :

(1) (a), (b) and (e) only
(2) (c), (d) and (e) only
(3) (b), (c) and (d) only
(4) (a), (b) and (c) only

31) Read the given statements and select the **correct** option.
Statement X : Part of nephron where maximum reabsorption of nutrients occur is lined by brush bordered columnar epithelium.

Statement Y : PCT helps to maintain the pH and ionic balance of the body fluids by selective secretion of hydrogen ions and potassium ions into the filtrate.

(1) Both the statements X and Y are correct
(2) Both the statements X and Y are incorrect
(3) Only statement X is correct
(4) Only statement Y is correct

32) Ultrafiltration in glomerulus occurs due to :-

(1) Bowman's capsule has higher hydrostatic pressure than glomerulus
(2) Colloidal osmotic pressure and capsule's hydrostatic pressure are lower than glomerular hydrostatic pressure
(3) Glomerulus hydrostatic pressure is less than capsular pressure
(4) Osmotic pressure is more than Hydrostatic pressure

33) Function of Henle's loop is

(1) passage of urine
(2) formation of urine
(3) conservation of water
(4) filtration of H_2O

34) Read the following statements carefully :-
(i) Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form myelin sheath around axon.
(ii) In a chemical synapse, pre & post-synaptic membrane in close preciosity.
(iii) Brain is the alternate information processing organ of our body.
(iv) These are two types of synapse-chemical & electrical synapses.
Choose the incorrect statement :-

(1) (ii) & (iv) only
(2) only (i), (iii) & (iv)
(3) (ii) & (iii)
(4) Only (iii)

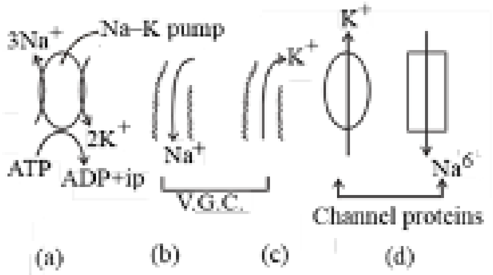
35)

Choose the correct option with reference of structure with their characteristic and location

	Structure	**Characteristic**	**Location**
(1)	Neuron	conductivity excitability	Myelinated nerve fibres are found in CNS formed by schwann cell
(2)	Midbrain	Consist of 4 optic lobes	Between forebrain and pons
(3)	Hypothalamus	Secretion of hormones	On ventral surface of cerebellum
(4)	Piamater	highly vascular layer	Around brain

(1) 1
(2) 2
(3) 3
(4) 4

36) Find out the answers of given statements according to diagrams :-



Which of above diagram/pathways are closed during depolarisation of neuron ?

(1) Only (a) & (d)
(2) Only (a) & (c)
(3) Only (a) & (b)
(4) Only (b) and (c)

37) Choose the correct and wrong statements from the following :-
a. Largest part of brain is cerebrum.
b. Collection of axon in PNS is called nuclei.
c. Mg^{+2} help in synaptic transmission.
d. Schwann cells help in myelinogenesis in PNS.
e. Typical value of RMP is – 70 mV.

	Correct	**Wrong**
(1)	a, b, c	d, e
(2)	a, d, e	b, c
(3)	a, c, d	b, e
(4)	a, d	b, c, e

(1) 1
(2) 2
(3) 3
(4) 4

38) Which of the following is incorrectly matched ?

(1) Crura cerebrai – Limb-muscle control
(2) Hypothalamus – Body temprature
(3) Cerebrum – vomiting centre
(4) Medulla oblongata – Respiratory centre

39)

Find out the correct match

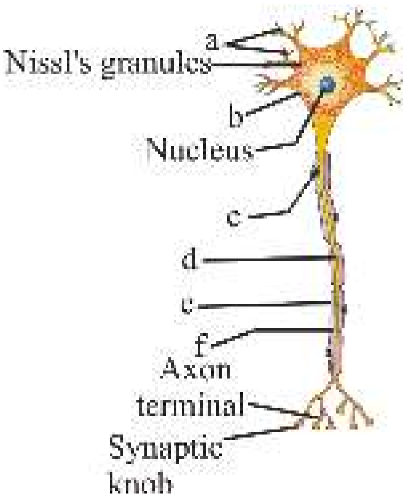
	Column-I		Column-II
A	Dendron	i	Protein synthesis
B	Nissl's granule	ii	Saltatory conduction
C	Neurofibril	iii	Afferent part
D	Myelinated Axon	iv	Efferent part
E	Nerve fibre	v	helps in internal conduction

(1) A-iii B-i C-ii D-iv E-v

(2) A-iv B-i C-ii D-iii E-v

(3) A-iv B-v C-i D-ii E-iii

(4) A-iii B-i C-v D-ii E-iv



40) Recognise the figure and find out the correct matching :-

(1) a-axons, b-cell body, c-myelin sheath, d-dendrites, e-node of Ranvier, f-Schwann cell

(2) a-dendrites, b-axon, c-Schwann cell, d-cell body, e-myelin sheath, f-node of Ranvier

(3) a-dendrites, b-cell body, c-Schwann cell, d-axon, e-myelin sheath, f-Node of Ranvier

(4) a- synaptic knobs, b-cell body, c-myelin sheath, d-axon, e-Schwann cell, f-Node of Ranvier

41) Myelin sheath of neuron is :-
(a) A layer around axon helps in saltatory conduction
(b) Made up of sphingolipids
(c) Synthesize by oligodendrocytes in PNS
(d) Is of conducting nature
(e) Is of non-conducting in nature

(1) Only a & b are incorrect

(2) Only a, b & e are correct

(3) All are incorrect
(4) a, b, c, e are incorrect

42) Functional and structural unit of nervous system :-

(1) Nephron
(2) Hepatic lobule
(3) Alveoli
(4) Neuron

43) Resting membrane potential is :

(1) –35 mv
(2) +70 mv
(3) –70 mv
(4) +35 mv

44) The PNS comprises of all the nerve of the body associated with ?

(1) ANS only
(2) Spinal cord only
(3) Brain only
(4) Both spinal cord and brain

45) Neuron with one axon and without dendron are normally found in :-

(1) Retina
(2) Embryonic stage
(3) Amacrine cell
(4) Cerebral cortex

46) Following statements are correct about nerve transmission except

(1) Depolarisation - Na^+ ions
(2) Repolarisation - K^+ ions
(3) Depolarisation - Na^+–K^+ pump closed
(4) Hyperpolarisation - Receiving of Stimulus

47) Out of the following, which tissue is originated by ectoderm?

(1) Connective tissue
(2) Areolar tissue
(3) Muscular tissue
(4) Nervous tissue

48) The iter/cerebral aqueduct lies :-

(1) In the third ventricle
(2) In the second ventricle
(3) Between the third and the fourth ventricles
(4) In the lateral ventricles

49) Division is not possible in neuron due to abscence of :-

(1) Mitochondria
(2) Centriole
(3) Nissl's granule
(4) Neurofibril

50) Pons varoli is characterised by a centre called :-

(1) Hunger centre
(2) Thirst centre
(3) Cardiac acceleratory centre
(4) Pneumotaxic centre

51) Parkinson's disease occur due to hyposecretion of :-

(1) Acetylcholine
(2) Dopamine
(3) GABA
(4) Sertonin

52) Central sulcus is found between

(1) Frontal and temporal lobe
(2) Frontal and occipital lobe
(3) Temporal and parietal lobe
(4) Parietal and frontal lobe

53) Multipolar neurons are found in the :-

(1) Retina of eye
(2) Embryonic stage
(3) Both (1) and (2)
(4) Cerebral cortex

54) In nerve conduction, during depolarisation :-

(1) Influx of Na^+ occur
(2) Outflux of K^+ occur
(3) Influx of Na^+, K^+ both occur

(4) Outflux of Na^+, K^+ both occur

55) Ventricles of brain are lined by the cells called-

(1) Neuroglia
(2) Ependymal
(3) Neurocells
(4) Schwann cells

56) Temporal lobe does not contain

(1) Wernicke's area
(2) olfactory area
(3) Auditory area
(4) Broca's area

57)

Which one of the following neuroglial cells helps in formation of blood brain barrier?

(1) Oligodendrocytes
(2) Astrocytes
(3) Microgliocyte
(4) Ependymal cell

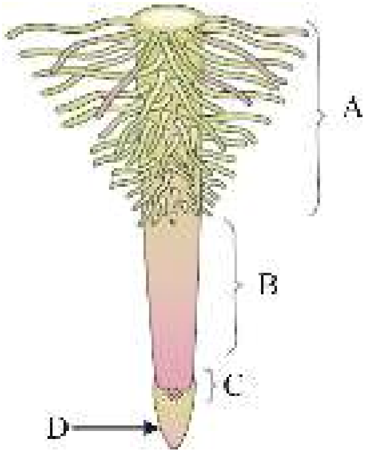
58) Find out the correct match from the following table :-

	Column-I	**Column-II**	**Column-III**
(i)	Drupe	Coconut	Edible endosperm
(ii)	Drupe	Mango	Edible fleshy aril
(iii)	True fruit	Mango	Fleshy meso carp

(1) i only
(2) i and iii
(3) iii only
(4) i, ii and iii

59) Mustard plant is developed from :-

(1) Perigynous flower
(2) Superior ovary
(3) Epigynous flower
(4) Bicarpallory ovary

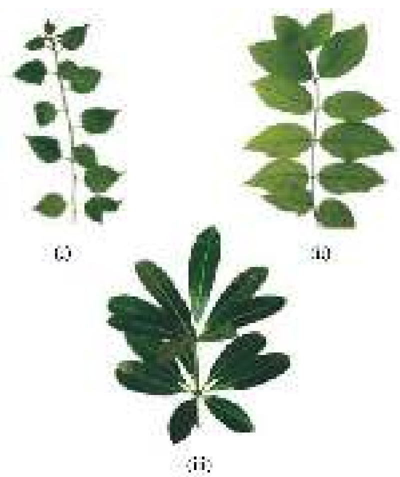
60) 

	Region of elongation	Region of Meristematic growth	Region of maturation
(1)	A	D	B
(2)	B	C	A
(3)	C	A	D
(4)	D	B	C

(1) 1
(2) 2
(3) 3
(4) 4

61)

Study the given figure and identify the kind of phyllotaxy :-



	(i)	(ii)	(iii)
(1)	Alternate	Opposite	Opposite
(2)	Alternate	Opposite	whorled
(3)	Opposite	Alternate	whorled

(4)	Opposite	whorled	Alternate

(1) 1
(2) 2
(3) 3
(4) 4

62) Prop roots of banyan tree are mainly for :-

(1) Respiration
(2) Absorption of water from soil
(3) retention of water in soil
(4) Providing support to large branches

63) Which of the following is incorrect?

(1) Root helps in water and mineral absorption from soil
(2) Root provide a proper anchorage
(3) Roots store food material and synthesise plant growth regulators
(4) Roots lack meristematic activity

64) **Statement-I :** When a flower can be divided into two equal radial halves in any radial plane, it is said to be zygomorphic.
Statement-II : When flower can be divided into two similar halves only in one particular vertical plane, it is actinomorphic.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II in correct.
(4) Both Statement-I & Statement-II are incorrect.

65) **Statement- I :** Brinjal is an example of hypogynous flower.
Statement- II : Brinjal is an example of inferior ovary.

(1) Both Statements I and II are correct
(2) Both Statements I and II are incorrect
(3) Only statement I is correct
(4) Only statement II is correct

66) Drupe fruits are found in :-

(1) Mango & Orange
(2) Mango & Apple
(3) Mango & Coconut
(4) Apple & Pear

67) Monocotyledonous seeds, consist of one large and shield shaped cotyledon known as:-

(1) Tegmen

(2) Endosperm

(3) Scutellum

(4) Aleurone

68) In a racemose inflorescence the main axis:

(1) Bear Solitary flower

(2) Terminates in a flower

(3) Has unlimited growth

(4) Has limited growth

69) Consider the following four statements (a-d) and select the option which includes all the correct ones only:-
(a) In Solanaceae family, flowers are zygomorphic
(b) In Solanaceae family, ten stamens are present in a flower
(c) In Solanaceae family, seeds are endospermic. (d) In Solanaceae family, axile placentation is present.

(1) Only statement (a) and (b)

(2) Only statement (b) and (c)

(3) Statement (a), (b), (c) and (d)

(4) Only statement (c) and (d)

70) Endospermous seeds, axile placentation, epipetalous stamens are found in:

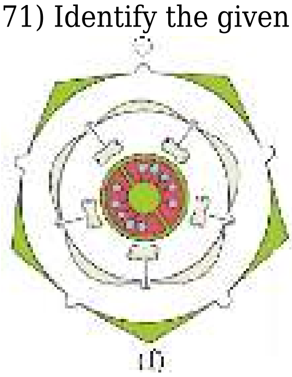
(1) Brassicaceae

(2) Fabaceae

(3) Liliaceae

(4) Solanaceae

71) Identify the given floral diagram and choose suitable floral formula from the following :-



(1) $\oplus$ ⚥ $K_{(5)} \; \overparen{C_{(5)} \; A_{(5)}} \; \underline{G}_{(2)}$

(2) $\oplus$ ⚥ $K_{(5)} \; \overparen{C_{(5)} \; A_{5}} \; \overline{G}_{(2)}$

(3) $\oplus$ ⚥ $K_{(5)} \; \overparen{C_{(5)} \; A_{5}} \; \underline{G}_{(2)}$

(4) $\oplus$ ⚥ $K_5 \ \overbrace{C_{(5)} \ A_{(5)}} \ \underline{G}_{(2)}$

72) When filaments of stamens are united and anthers are free, the condition is called as........and found in family..... :

(1) Epipetalous, Liliaceae.

(2) Epitepalous, Fabaceae

(3) Syngenesious, Asteraceae

(4) Adelphy, Malvaceae

73) How many statements are incorrect with respect to given floral formula

Br $\oplus$ ⚥ $K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$

(a) Flower has bicarpellary apocarpous condition
(b) Flower is Actinomorphic & bisexual
(c) Four petals with gamopetalous condition
(d) Stamens are forming two bundle and its called diadelphy
(e) Flower is hypogynous

(1) Two

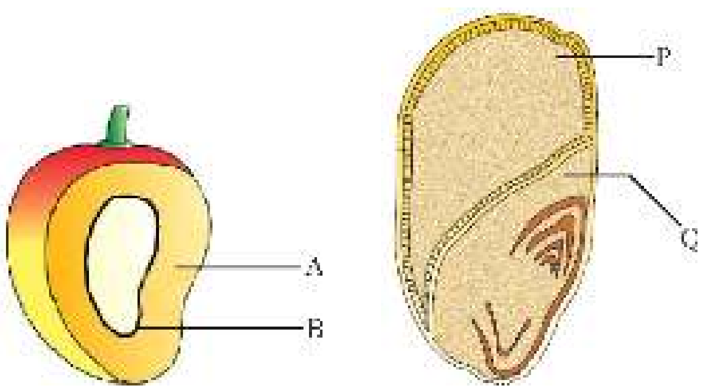
(2) Three

(3) Four

(4) Five

74)

Identify the parts labelled A,B and P,Q in the figure given below and select the right option.



	A	B	P	Q
(1)	Epicarp	Mesocarp	Scutellum	Endosperm
(2)	Mesocarp	Endocarp	Endosperm	Scutellum
(3)	Aleurone layer	Endocarp	Mesocarp	Radicle
(4)	Mesocarp	Seed	Endosperm	Plumule

(1) 1

(2) 2

(3) 3

(4) 4

75) Number of fatty acids present in a molecule of phospholipid is :-

(1) Two
(2) Three
(3) one
(4) none of these

76) Each globin polypeptide chain in haemoglobin molecule has :-

(1) Primary structure
(2) Secondary structure
(3) Tertiary structure
(4) Quaternary structure

77) Select odd one :-

(1) Chitin – Homopolysaccharide
(2) Hyaluronic Acid – Heteropolysaccharide
(3) Carotenoid – Secondary metabolites
(4) Wax – Conjugated lipid

78) A nucleoside differs from a nucleotide is not having :-

(1) Sugar
(2) Phosphate
(3) Nitrogen base
(4) Phosphate & sugar

79) **Assertion (A) :** 3-dimensional view of protein is indicated by a tertiary structure.
Reason (R) : Hydrogen bond, Disulphide bond hold two or more protein molecule together.

(1) (A) is true but (R) is false
(2) Both (A) & (R) are false
(3) Both (A) & (R) are true but (R) is not the correct explanation of (A).
(4) Both (A) & (R) are true but (R) is the correct explanation of (A).

80) Read the following statements :
(a) 20 carbon atoms are present including carboxyl carbon.
(b) Unsaturated fatty acids with one or more double bond.
(c) Polyunsaturated fatty acid.
Which of the following lipid is being described by above statements ?

(1) Arachidic acid
(2) Arachidonic acid
(3) Palmitic acid

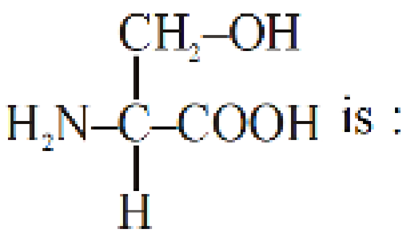
(4) Oleic acid

81) **Assertion (A) :** At high temperature enzymatic activity is destroyed.
Reason (R) : Proteins are denatured by heat.

(1) Both (A) & (R) are true but (R) is the correct explanation of (A).
(2) Both (A) & (R) are false
(3) (A) is false but (R) is true
(4) Both (A) & (R) are true but (R) is not the correct explanation of (A).

82) Lecithin is a :-

(1) Sterol
(2) Glycolipid
(3) Phospholipid
(4) Sphingolipid

83) 

(1) Non-essential AA
(2) Serine
(3) Tyrosine
(4) Both (1) and (2)

84) Match the column I with column II and choose correct option :-

	Column-I		Column-II
i	lectins	a	Codeine
ii	Essential oil	b	Lemon grass oil
iii	Pigment	c	Concanavalin A
iv	Alkaloid	d	Anthocyanin

	i	ii	iii	iv
1.	d	c	b	a
2.	c	b	d	a
3.	a	b	d	c
4.	b	c	d	a

(1) 1

(2) 2

(3) 3

(4) 4

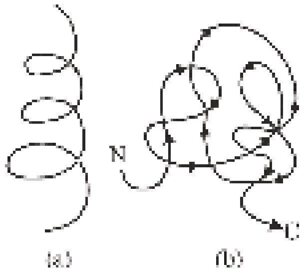
85) In starch molecule, the types of bonds which exist in the backbone and at the site of branching are respectively :-

(1) α-1, 2-glycosidic bond and α-1, 4-glycosidic bond

(2) β-1, 4-glycosidic bond and β-1, 6-glycosidic bond

(3) α-1, 4-glycosidic bond and α-1, 6-glycosidic bond

(4) α-1, 2-glycosidic bond and α-1, 6-glycosidic bond



86) Recognise the figure and find out the correct matching.

(1) a-Primary structure   b-Secondary structure

(2) a-Secondary structure   b-Primary structure

(3) a-Secondary structure   b-Tertiary structure

(4) a-Tertiary structure   b-Quaternary structure

87) Which of the following statement is wrong :-

(1) Glycerol is trihydroxy propane.

(2) Starch gives blue colour in iodine test.

(3) Lipids are strictly macro biomolecules.

(4) Starch is homopolysaccharide

88) Most abundant element in human body and least abundant element in earth crust are respectively :-

(1) O and Si

(2) O and N

(3) O and S

(4) O and C

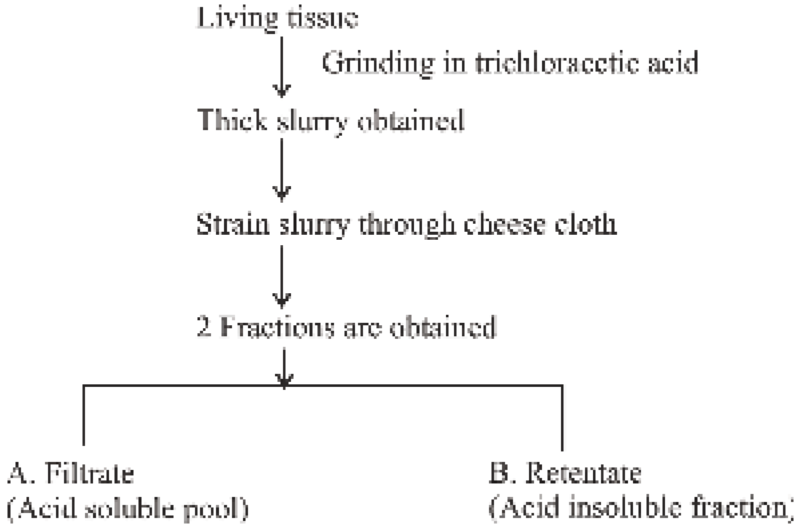
89) Read the following statements and find out which is/are **correct** ?
(A) Proteins are homopolymers of amino acids.
(B) Amino acids are substituted ethanes.
(C) Glycerol is also known as trihydroxy ethane.
(D) Paper made from plant pulp is cellulose.
(E) RuBisCO is most abundant protein in animals.

(1) A, B, C and E

(2) B, C and D

(3) A and E

(4) Only D



90)
I. Molecular weight ranging from less than 1000 Dalton approximately
II. Has four type of organic compounds proteins, nucleic acids, polysaccharides and lipids
III. Contain chemicals that have molecular weight more than 1000 Dalton
IV. Has monomers
V. Has generally polymers except lipid
VI. Mostly micromolecules
VII. Generally macromolecules
How many of the above statements (I-VII) are correct regarding A & B respectively ?

	A	**B**
(1)	I, II, III	IV, V, VI, VII
(2)	II, IV, VI	I, III, V, VII
(3)	I, IV, VI	II, III, V, VII
(4)	I, III, V	II, IV, VI, VII

(1) 1

(2) 2

(3) 3

(4) 4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**4**	**2**	**4**	**3**	**1**	**3**	**3**	**2**	**4**	**4**	**3**	**3**	**1**	**4**	**3**	**3**	**2**	**2**	**1**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**1**	**4**	**3**	**3**	**3**	**3**	**4**	**1**	**1**	**3**	**4**	**3**	**4**	**1**	**3**	**1**	**1**	**3**	**3**	**2**
Q.	41	42	43	44	45															
A.	**3**	**2**	**2**	**3**	**2**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**2**	**3**	**4**	**1**	**2**	**4**	**2**	**3**	**2**	**2**	**3**	**2**	**1**	**2**	**1**	**4**	**3**	**4**	**1**	**3**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**2**	**2**	**4**	**3**	**2**	**3**	**2**	**1**	**4**	**2**	**2**	**3**	**4**	**3**	**4**	**2**	**3**	**1**	**1**	**3**
Q.	86	87	88	89	90															
A.	**2**	**4**	**1**	**1**	**1**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**2**	**3**	**3**	**4**	**3**	**2**	**2**	**1**	**3**	**2**	**2**	**2**	**4**	**2**	**3**	**4**	**3**	**3**	**2**	**2**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**3**	**2**	**1**	**3**	**3**	**3**	**4**	**4**	**3**	**4**	**4**	**2**	**3**	**3**	**2**	**2**	**2**	**3**	**4**	**3**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**2**	**4**	**3**	**4**	**2**	**4**	**4**	**3**	**2**	**4**	**2**	**4**	**4**	**1**	**2**	**4**	**2**	**2**	**2**	**2**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**2**	**4**	**4**	**4**	**3**	**3**	**3**	**3**	**4**	**4**	**3**	**4**	**2**	**2**	**1**	**3**	**4**	**2**	**1**	**2**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**1**	**3**	**4**	**2**	**3**	**3**	**3**	**2**	**4**	**3**										

SOLUTIONS

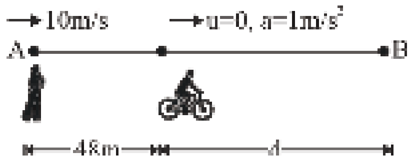
PHYSICS

1) Let the velocities of two trains be u and v, such that u > v.
Relative velocity while crossing = u + v
Relative velocity while overtaking = u – v

Given : $10 = \frac{400}{u+v}$

and $20 = \frac{400}{u-v}$

Solving we get, u = 30 ms^{-1} and v = 10 ms^{-1}



2)
Let the boy meet the cyclist at P.
Boy (uniform motion) : 10 t = 48 +d ...(i)

Cyclist (accelerated motion: d = $\frac{1}{2}$ .1.t^2) ...(ii)

$10t = 48 + \frac{t^2}{2} \Rightarrow t^2 - 20t + 96 = 0$

(t – 8) (t – 12) = 0
The boy meets the cyclist at t =8 s, 12s
At t = 8s, velocity of cyclist, ν_c=1 × 8 = 8 m/s
$\nu_b > \nu_c$, boy crosses the cyclist
at t = 12 sec, $\nu_c > \nu_b$, cyclist crosses the boy and will be always ahead

3) $T \propto v \; R \propto u^2$

4) $\frac{H}{R} = \frac{\tan\theta}{4}$

$\frac{H}{4H} = \frac{\tan\theta}{4}$

tanθ = 1
⇒ θ = 45°

5)

$u_y = 3$, $u_x = 4$
Range = $u_x t$
$-36 = 3t - 5t^2 \Rightarrow t = 3s$
Range = 4 × 3 = 12 m

6) $V_{rm} = \sqrt{v_r^2 + V_m^2}$

$= \sqrt{(10)^2 + (3)^2}$

$= \sqrt{109}$

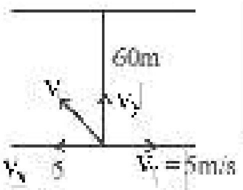
7) $v_{RG} = \sqrt{v_R^2 + v_G^2}$

$v_{RG} = \sqrt{(4)^2 + (3)^2}$ = 5 km/h

8) $v = \sqrt{5^2 + 12^2}$

$\sqrt{169}$ 13m/s

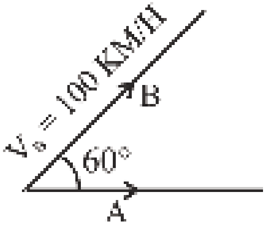
$v_y = \frac{60}{5} = 12\,m/s$



9) $t = \frac{d}{v_{SR}} = \frac{500}{3 \times 5} \times 18 = 600$ sec

$t = \frac{600}{60} = 10\,\text{min}$

10) $t = \frac{t_1 t_2}{t_1 + t_2} = \frac{60 \times 40}{100} = 24\,\text{sec}$



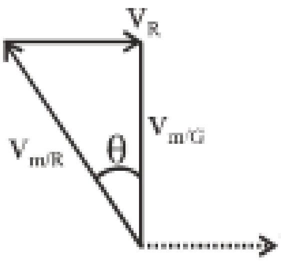
11) V_A = 50 KM/H

$\vec{V}_A = 50\,\hat{i}$

$\vec{V}_B = 50\hat{i} + 50\sqrt{3}\hat{j}$

$\vec{V}_{BA} = 50\sqrt{3}\hat{j}$

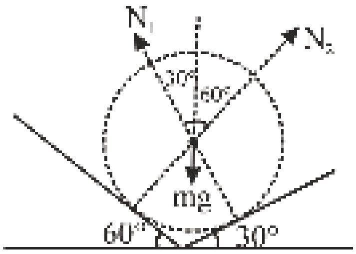
$\left|\vec{V}_{BA}\right| = 50\sqrt{3}$ KM/Hr

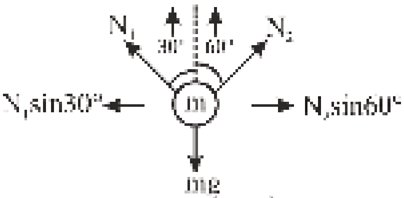
12) 

sin θ = $\frac{v_R}{v_{m/R}} = \frac{v_R}{nv_R} = \frac{1}{n}$

θ = $\sin^{-1}\left(\frac{1}{n}\right)$

from river current = $\frac{\pi}{2} + \sin^{-1}\left(\frac{1}{n}\right)$

13) 



$N_1\left(\frac{1}{2}\right) = N_2\left(\frac{\sqrt{3}}{2}\right)$

$\boxed{N_1 = \sqrt{3}\,N_2}$

14) $F\cos 30° - T_1 = 15a$

$T_1 - T_2 = 12a$

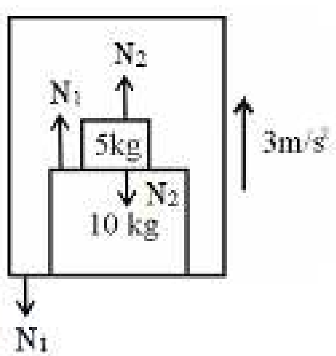
$T_2 = 3a$

Solving,

$T_1 = \frac{15}{30}.F\cos 30$

$T_2 = \frac{3}{30}.F\cos 30$

So, $\frac{T_1}{T_2} = 5$



15)

$N_1 = (10 + 5)(g + a)$

$N_1 = 15(10 + 3)$

$N_1 = 195$ N

16) Downward force = mg sinθ. With decreases in value of θ, this downward force also decreases.

17) **1. Asking About**
We are tasked with determining the speed of a 5 kg object at x = 25 m, given that it starts from rest (v = 0) at x = 0. The force acting on the object varies with its position, as shown in the graph described.
2. Concept
• **Work-Energy Theorem:** The total work done on an object is equal to its change in kinetic energy:
$W = \Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2.$
Since the object starts from rest (v^0 = 0), the equation simplifies to: $W = \frac{1}{2}mv^2.$
• The work done by the force is the **area under the Force vs. Position graph** from x = 0 to x = 25.
3. Formula
1. Work done (W): W = Area under the Force-Position graph.
2. Kinetic energy: $W = \frac{1}{2}mv^2.$
3. Speed: $v = \sqrt{\frac{2W}{m}}.$
4. Solution/Explanation
1. Analyze the graph:
• From x = 0 to x = 25m the force is constant at F = 10N.
• The area under this part of the graph is a rectangle: $\text{Area} = \text{Base} \cdot \text{Height}.$
2. Calculate work done (W):
• Base = 25 m
• Height = 10 N.
$W = 25 \cdot 10 = 250\text{J}.$
3. Apply the work-energy theorem:
• Using $W = \frac{1}{2}mv^2$, $250 = \frac{1}{2} \cdot 5 \cdot v^2.$
4. Solve for v: $250 = 2.5v^2,$
$v^2 = \frac{250}{2.5} = 100$,
$v = \sqrt{100} = 10\text{m/s}.$
5. Final Answer
The speed of the object at x = 25 m is:
$\boxed{10\text{m/s}}.$
6. Correct Option: 2

18) $\vec{F} = \frac{d\vec{p}}{dt} = m\frac{d\vec{v}}{dt}$ where $\vec{v} = \frac{d\vec{r}}{dt} = 8t\hat{i} + 2\hat{j} - 6\hat{k}$ So $\vec{F} = 2\left[8\hat{i}\right] = 16\hat{i}$

19) When the lift is accelerating upwards with a constant acceleration a, then apparent weight,
w = m(g + a) = 60 (10 + 2) = 60 × 12 = 720 N
when the lift is accelerating downwards at the rate a, then apparent weight,
w = m(g – a)
= 60(10 – 2) = 60 × 8 = 480 N

20) $\text{Force} = \frac{\Delta\vec{p}}{\Delta t} = \frac{m(\vec{v}-\vec{u})}{\Delta t}$
$-60 = \frac{1(0-u)}{0.1} \Rightarrow u = 6\,m/s$

21) $a = \frac{10^4}{10 \times 1000} = 1\ m/s^2$
$F = 4 \times 1000 \times 1 = 4 \times 10^3$ N

22) Momentum acquired by the particle is numerically equal to the area enclosed between the F-t curve and time Axis. For the given diagram area in a upper half is positive and in lower half is negative (and equal to the upper half.) So net area is zero. Hence the momentum acquired by the particle will be zero.

23) $a = \frac{F}{m} = \frac{20}{2} = 10\ m/s^2$
$v = u + at = 10 + 10 \times 5 = 60$ m/s

24) $\frac{dp}{dt}$

25) **Explanation :**
We are asked to determine which concept is connected by Newton's Second Law of Motion.
Concept :
Newton's Second Law of Motion states that :
F = dP/dt
Where :
• F is the net external force acting on an object
• dp/dt is the rate of change of momentum of the object.
This law essentially links the force acting on an object with how its momentum changes over time.
Formula :
The key formula for Newton's Second Law of Motion is
F = dP/dt
Where :
• P is momentum, defined as P = m.v, with m being the mass of the object and v its velocity
Calculation :
• Newton's Second Law specifically connects the rate rate of change of momentum with the external force acting on an object.
This, the correct answer is option 3, Rate of change of momentum and external force.

26) **1. Explanation:**
When you push an object (like a book), according to Newton's third law, the book exerts an equal and opposite force back on you. This means that the force the book exerts on you will be equal in magnitude but opposite in direction to the force you apply.
2. Concept:
Newton's third law states:

3. Given Force:

You apply the forcer

F - $3\hat{i} + 4\hat{j}$N

4. Calculation:

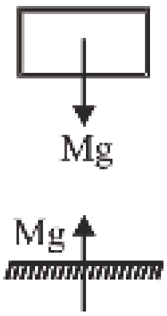
The force the book exerts on you will have the same magnitude but the opposite direction:

$F_{book} = -(3\hat{i} + 4\hat{j}) = -3\hat{i} - 4\hat{j}$ N

Final Answer:

The force exerted by the book on you is

$F_{book} = -3\hat{i} - 4\hat{j}$ N

27) 

28)

Explanation :

Ask about action – reaction law.

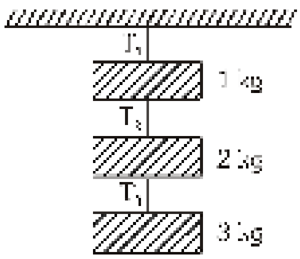
Concept :

Newton’s twird law of motion. According to this law every mutual interaction between two bodies always follow action- reaction law no single isolated force is not possible without reaction.

29)

By theory

30)
$T_2 = (2 + 3)g$
$= 5g$
$=50N$



31)

Explain :

analyze two statements about action-reaction pairs and their relationship with weight and normal reaction on a block placed on a horizontal surface.

Concept :
Action-reaction pairs (Newton's third law) involve forces between two interacting objects, while weight and normal reaction, in this scenario, act on the same object.

Calculation :
Weight is the gravitational force exerted by the Earth on the block. The reaction force to the weight is the gravitational force exerted by the block on the Earth.
Normal reaction is the force exerted by the surface on the block, preventing it from passing through the surface. The reaction force to the normal reaction is the force exerted by the block on the surface.
Weight and normal reaction act on the same object (the block), which is contrary to the action-reaction pair definition.
Action-reaction pairs (Newton's third law) always involve two different objects.
For instance, if object A exerts a force on object B (action), then object B exerts an equal and opposite force on object A (reaction).
Statement-1 and 2 both are false.

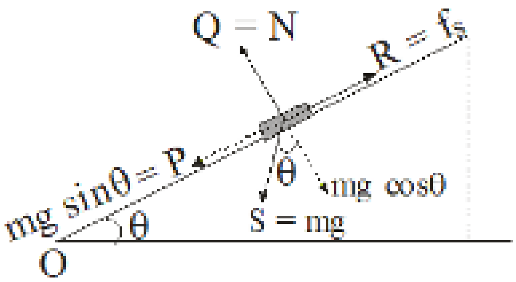
Ans (4) Difficulty level : easy

32) Conceptual.

33)

Force of gravity acts always vertically downward.

34) According to law of inertia (Newton's first law), when cloth is pulled from a table, the cloth come in state of motion but dishes remains stationary due to inertia. Therefore when we pull the cloth suddenly from table the dishes remains stationary.

35) 

Q = Normal reaction
P = mg sinθ
R = Friction force
S = Weight (mg)
(a)-(ii), (b)-(iii), (c)-(iv) and (d)-(i)

36)

I = Area of F-t graph

37) **Solution Explanation :**
To find the train in speeding up or slow down.
Concept :
Law of inertia.
Formula :
Not required
Calculation : When the coin is tossed, it intially two the same horizontal velocity as the train but if the train acceleration, the person moves forward along with the train, while the coin, which is no longer influenced by the train accleration, lags behind and appears to fall behind the person.
Answer : 1
Level Easy

38) **Question Explanation (30 words)** A 10 N force is applied at a 37º angle on a block placed on a smooth horizontal surface. We need to determine the block's acceleration using given values of trigonometric functions. **Concept Based (30 words)**
The problem is based on Newton's Second Law of Motion and force resolution. The applied force is resolved into horizontal and vertical components to determine the horizontal acceleration of the block.
Formula Used :

A. **Force resolution:**

$F_x = F \cos\theta$
$F_y = F \sin\theta$

A. **Newton's Second Law:**

$F_x = ma$

Calculation
Given:
$F = 10N$, $\theta = 37º$, $\sin 37º = \frac{3}{5}$, $\cos 37º = \frac{4}{5}$
$g = 10\ m/s^2$, Assume $m = 1$ kg

Resolving forces:
$F_x = 10 \cos 37º = 10 \times \frac{4}{5} = 8N$
$F_y = 10 \sin 37º = 10 \times \frac{3}{5} = 6N$
Since the surface is smooth, no vertical acceleration occurs.
Using Newton's Second Law:
$F_x = ma$
$8 = 1 \times a$
$a = 8\ m / s^2$
Correct answer : 3

39)

$v = \frac{dx}{dt} = 3t^2 - 75$

$a = \frac{dv}{dt} = 6t$

F = ma = 6 × 6 × t

at t = 4, F = 36 × 4 = 144 newton

40)

$v\frac{dm}{dt} = mg$

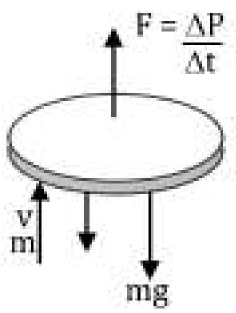
$\Rightarrow \frac{dm}{dt} = \frac{mg}{v} = \frac{500 \times 10}{1000}$ = 5 kg/sec

41) **Explain:** A disc is kept floating in the air by firing bullets vertically at it. Find the speed with which the bullets are fired, given the mass of the disc, the mass and rate of firing of the bullets, and the fact that the bullets rebound with the same speed.

Concept: For the disc to float, the upward force exerted by the rebounding bullets must balance the weigh of the disc.

Formula: Force = Rate of change of momentum = $\frac{\Delta P}{\Delta t}$

Momentum P = mv



Calculation: The change in momentum of each bullet during collision is :

$\overrightarrow{\Delta P} = \overrightarrow{P}_{final} - P_{initial}$ = (–mv) – (mv) = –2mv (downwards)

The magnitude of the change in momentum = |-2mv| = 2mv.

The force exerted by each bullet on the disc is the rate of change of momentum of the bullet during the collision.

In one second, n = 10 bullets are fired.

The total change in momentum of the bullets per second = Total $\overrightarrow{\Delta P}$ per second = n × | –2mv| = 2mnv

For the disc to float, this upward force must balance the weight of the disc: Upward force = Weight of the disc

2mnv = Mg

$v = \frac{Mg}{2mn} = \frac{1.9.8}{2 \times 0.05 \times 10}$

Ans. (3)

Difficulty level : tough

42) **Solution:**

Asking About:
Find the maximum acceleration (a) of the stone without breaking the cord.

Concept:

A. **Newton's Second Law:** F = ma
B. **Tension Limit:** The maximum tension the cord can withstand is three times the weight of the stone i.e., $T_{max} = 3mg$.

Formula:
$T_{max} = m(g + a)$

Calculation / Explanation:
$3mg = m(g + a)$
$3g = g + a$
$a = 2a$

Final Answer: 2g
Correct option is (2).

43) Newton second law

$F = ma \Rightarrow 6 = (7 + 5)a;\ a = \frac{1}{2}\ m/s^2;$

Now, $F' = 5 \times \frac{1}{2} = 2.5$ N

44) $F = v\frac{dm}{dt} = (5 \times 10^4)(100) = 5 \times 10^6$ N

45) $T_1 \sin 30 + T_2 \sin 60 = 50$
$T_1 \cos 30 = T_2 \cos 60$
$T_1 + \sqrt{3}T_2 = 100$ ....(i)
$\sqrt{3}\ T_1 = T_2$ ....(ii)
From (i) & (ii)
$T_1 = 25$ N
$T_2 = 25\sqrt{3}$ N

CHEMISTRY

46) $O_2(g) \rightleftharpoons CO_2(g)$,
n_o : 4 3
n_e : 4+x 3–x (∵ $Q_p > K_p$)
∴ $= K_p = \frac{3-x}{4+x}$ $K_p = \frac{1}{5}$

47) $P_4(g) + 6Cl_2(g) \rightleftharpoons 4PCl_3(g)$
t = 0 xM xM 0

At equilibrium

(x–y)M (x–6y)M 4xM

$[P_4] > [Cl_2]$

48) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

Let initial conc. 2 0 0

at eq^m 2-1 0.5 0.5

$$K_C = \frac{0.5 \times 0.5}{1} = 0.25$$

49) High pressure will favour the reaction to proceed in direction where number of moles of gases component decreases.

50) $$K_c = \frac{n_B . n_C^3}{n_A^2} \times \frac{1}{V^2}$$

$$\Rightarrow 16 = \frac{2 \times 2^3}{2^2 \times V^2}$$

$$\Rightarrow V = \frac{1}{2}$$

51)

$A(g) + B(g) \rightleftharpoons 2C(g)$

3 1 0

3 – x 1 – x 2x

2x = 1.5

x = 0.75

2.25 0.25 1.5

$$K_C = \frac{\left(\frac{1.5}{1}\right)^2}{\left(\frac{2.25}{1}\right)\left(\frac{0.25}{1}\right)} = \frac{1}{0.25} = 4$$

52) $\frac{K}{Q} = 0.33 \Rightarrow \frac{K}{Q} < 1 \Rightarrow K < Q$ Reaction proceed in net backward direction

53)

As per theory

54) **<u>Explanation</u>:-** Find Kp

<u>Given data</u>:- $(P_T)_{eq}$ = 0.6 atm

<u>Concept</u>:- Reactant A is solid

$A(s) \rightleftharpoons B(g) + C_{(g)}$

t eq p p

0.3 0.3

$P_T \Rightarrow 2P = 0.6$
P = 0.3 atm
<u>Formula</u>:- $kp = (P_B)(P_C)$
<u>Calculation</u>:- $Kp = 0 \cdot 3 \times 0 \cdot 3 = 0 \cdot 09$
<u>Final Answer</u>:- 0.09
Correct Answer:- (2)

55)

Reverse the reaction, inverse the equilibrium constant and if a reaction is multiplied by some number then that number goes to power of old equilibrium constant.

$$A \rightleftharpoons B \qquad K_c = 2$$

$$2B \rightleftharpoons 2A \qquad K_c = \frac{1}{4} \qquad ...(i)$$

$$B \rightleftharpoons C \qquad K_c = 0.01$$

$$2C \rightleftharpoons 2B \qquad K_c = 10^4 \qquad ...(ii)$$

$$(i) + (ii)$$

56) $2C \rightleftharpoons 2A = 2.5 \times 10^3$

57)

As per theory

58)

$Ti^{2+} < Ti^{3+} < Ti^{4+}$
Greater the positive charge in same element greater will be Z_{eff} and less will be the radius.

59) 120 = $[U_{40}]\ 8s^2$
Therefore it can be placed in 8^{th} period and 2^{nd} group.

60) After losing 1 e^-, Oxygen shows most stable electronic configuration.

61)

Question Explanation: Identify correct option for given assertion & reason.
Concept: This question based on Electronic configuration.
Solution: Both Assertion and reason are correct – elements in a group show similar chemical properties.

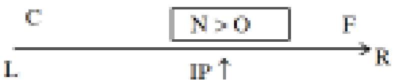
Reason is the correct explanation: This similarity is due to their similar outermost electronic configuration.
Final Answer: (4)

62)

With increase in atomic number atomic radius decreases along a period.

63) Correction were done in atomic weight of elements are U, Be, In, Au, Pt

64) NCERT XIth Pg.#85 Para-3 (Part-I)

65) 

IP ⇒ F > N > O > C
NCERT - XI, Part-I, Page No. 88, Edition 20-21, figure 3.6

66)

Among isoelectronic species,
ionic radii $\propto \frac{1}{Z_{eff}}$

67) **Question Explanation:-** The atomic number 34 corresponds to the element Selenium (Se).
Concept:- Electronic configuration
Se = $1s^2 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^4$
Final Answer: Option (2). 4^{th} period, 16^{th} group, p block.

68)

Explanation:
The question asks which orbital is filled immediately after the np orbital in an atom, following the Aufbau principle and the (n+l) rule for determining electron filling order.

Concept: (Aufbau principle & e-filling)
According to the Aufbau principle and the (n+l) rule, after the np orbital is filled, the next orbital to receive electrons is the (n+1)s orbital. This is due to its lower energy.

Final Answer : Option (4)

69)

Group no. for d-block.
(n – 1) d + ns e^-

5 + 2 = 7 group.

70) $Pb > Pb^{+2} > Pb^{+4}$ $\left[Z_{eff} \propto \frac{\oplus charge}{\ominus charge} \propto \frac{1}{size}\right]$

71) **Question Explanation :** Identify the set of elements with very similar atomic radii.
Concept: This question is based on Atomic radius trend.
Solution: Fe, Co, and Ni are consecutive transition metals. The increasing nuclear charge is offset by the screening effect of the d-electrons, resulting in a nearly constant atomic radius.
Final Answer: Option (3)

72) 4d e^- start from 39 elements.
Elements having 4d e^- = 100 – 38 = 62

73)

(1)	$(n-2)f^{14}(n-1)d^4\ ns^2$	(c)	Transition element
(2)	Ga	(d)	Eka aluminium
(3)	Ar	(a)	Noble gas
(4)	Cl	(b)	Halogen

74) **Explanation:**
Question is asking about element having highest Z effective.
Concept:
Z effective of Isoelectronic species depends directly on +ve charge
Solution:
$H^-Li^+Be^{+2}B^{+3}$ are all isoelectronic species having 2 electrons so B + 3 will have highest Z effective.
Answer: 4

75) $A\ell \simeq Ga$ Transitional contraction.

76)

18th group

77) **Question Explanation :**
Question is asking about comparison of size between Mo and W
Concept : Comparison of size of elements.

Explanation & Solution :
The atomic radii of Molybdenum (Mo) and Tungsten (W) are very similar primarily due to lanthanide contraction. This effect causes a decrease in the atomic size of elements in the 6th period (where W is located) compared to what would be expected based on their position in

the periodic table.

Answer- Option 3

78) Element Uut belongs to group 13 (p-block).

79) $Z_{eff} \propto \frac{+ve}{-ve}$

80) N = $2p^3$
= $\boxed{\upharpoonleft}\boxed{\upharpoonleft}\boxed{\upharpoonleft}$
3 unpaired

81) Conceptual

82)

Explanation:
The question asks you to identify the combination of statements that are correct about the periodic table and the elements within it.

Concept: (Modern periodic table & it's cocenpts, element positions)
(a) is incorrect; the modern periodic table is based on atomic number.
(b) is correct; the first d-block series ranges from Sc to Zn.
(c) is correct; alkali and alkaline earth metals are s-block elements.
(d) is correct; chalcogens and halogens are p-block elements.

Therefore, the correct statements are (b), (c), and (d).

Final Answer : Option (3)

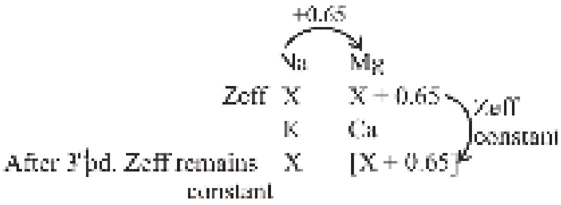
83)

$X^{+2}[Ar]3d^8$ = 26 'e'
hence X^0 = 28 'e'
$Y^{+3}[Ar]3d^3$ = 21 'e'
hence Y^0 = 24 'e'

84)

V.W.R □ 2 × Covalent Radius

85) Z_{eff} of Na = x



86) No. of valence electron = No. of IP before large difference of successine IP

87)

Due to transition contraction

88)

	Cl	
Se	Br	Kr
Te	I	Xe

89) Atomic no. Element Block 15 P p-Block
19 K s-Block
25 Mn d-Block
64 Gd f-Block

90) $_{58}Ce = {}_{54}[Xe]4f^2\ 6s^2$. f-block element placed in IIIB group.

BIOLOGY

91) **Explanation**
The given organisms belong to different orders of Class Mammalia:
1. Ornithorhynchus (*Platypus*) → Order Monotremata

A. Egg-laying mammal (Prototheria).

2. Macropus (*Kangaroo*) → Order Marsupialia

A. Marsupial (Metatheria).

3. Pteropus (*Flying Fox/Bat*) → Order Chiroptera

A. Placental mammal (Eutheria).

4. Balaenoptera (*Blue Whale*) → Order Cetacea

A. Largest marine mammal on Earth.
B. Completely aquatic with a streamlined body.

Correct Answer:
(2) Ornithorhynchus, Macropus, Pteropus & Balaenoptera

92)

NCERT-XI, Pg. # 40, Fig-4.4

93)

NCERT-XI, Pg. # 41, 47, 49

94)

NCERT (XI) Pg. # 57

95)

Explain Question : The structure given below (Diagram) is Present in which Animal ?

Concept : This question is based on Phylum Coelenterata.

Solution : The given Diagram is of Cnidoblast and Present in members of phylum Coelenterata
Eg. → Physalia

Final Answer : option (3)

96)

NCERT-XI, Pg. # 43

97)

NCERT-XI, Pg. # 42-50 Example of phylum.

98)

NCERT-XI, Pg. # 40

99) NCERT, Pg. # 38, 39

100)

NCERT-XI, Pg. # 40

101) NCERT -XI, Pg. # 42

102) NCERT (XI) Pg# 58 Para: 4.2.11.5

103)

Explain Question : Which is not a sponge ?

Concept : This question is based on Phylum porifera

Solution : option (4) Sea anemone is a Coelenterate. Rest all all sponge.

Final Answer : option (4)

104)

Explain Question : Identify the animal group in which coxal glands act as excretory organs.
Concept : This question is based on Arthropoda
Explanation : The correct answer is option 2.
Scorpion → Coxal glands -Correct
Cockroach → Malpighian tubules
Prawn → Green glands (antennal glands)
Leech → Nephridia
Final answer : 2

105)

Explain Question: Chitinous exoskeleton is found which phylum?
Concept: This question is based on Phylum Arthropoda.
Solution: Chitinous exoskeleton is found in members of phylum Arthropoda.
Final Answer: 3

106) **Explain Question:** Out of following in which phylum only sexual Reproduction occurs.
Concept: This question is based on Phylum ctenophore
Solution: ctenophore have only sexual Reproduction.
Final Answer: (4)

107)

NCERT-XI, Pg. # 39

108)

NCERT-XI, Pg. # 38

109)

NCERT-XI, Pg. # 44

110)

NCERT-XI, Pg. # 41

111)

NCERT-XI, Pg. # 44-45

112)

NCERT-XI, Pg. # 37

113)

NCERT-XI, Pg. # 45

114)

NCERT-XI, Pg. # 40

115)

NCERT Pg # 235

116)

NCERT-XI, Pg. # 213

117)

NCERT-XI, Pg. # 209

118)

NCERT-XI, Pg. # 212

119)

Solution :- Question belong zoology not my subject
Answer :- 3

120) **(a) Angiotensin II activates the cortex of adrenal gland to release aldosterone.**

A. This is **correct.** Angiotensin II stimulates the adrenal cortex to release **aldosterone,** which helps in regulating blood pressure by increasing sodium and water reabsorption in the kidneys.

(b) Aldosterone leads to an increase in blood pressure.

A. This is **correct.** Aldosterone increases sodium reabsorption in the kidneys, leading to water retention, which increases blood volume and, consequently, blood pressure.

(c) ANF acts as a check on renin-angiotensin mechanism.

A. This is **correct. Atrial Natriuretic Factor (ANF)** inhibits the renin-angiotensin system, leading to vasodilation and reduced blood pressure.

121)

NCERT-XI, Pg. # 209

122)

Explain Question : Ultrafiltration in glomerulus occurs in ?

Concept : This question is based on Mechanism of urine formation.

Solution : option (2) is correct
Blood colloidal osmotic pressure + Capsular Hydrostatic pressure are lower than glomerular Hydrostatic pressure
Net filtration = BGHP – (BCOP + CHP)
= 10-25 mg

Final Answer : option (2)

123) NCERT (XIth) Pg. # 295

124) NCERT Pg. (E) # 317, 318 ; (H) # 316, 317

125)

NCERT-XI, Pg. # 236

126)

NCERT-XI, Pg. # 232

127)

NCERT-XI, Pg. # 235

128)

Nurture Module No. 6

129)

NCERT-XI, Pg. # 231, 232

130)

The correct answer is **3.**
a-dendrites, b-cell body, c-Schwann cell, d-axon, e-myelin sheath, f-Node of Ranvier
Let's break down the components of the neuron:
- **Dendrites (a):** These are short, branching fibers that receive signals from other neurons.
- **Cell Body (b):** This is the main part of the neuron, containing the nucleus and other organelles.
- **Schwann Cell (c):** These cells wrap around the axon, forming the myelin sheath.
- **Axon (d):** This is a long, slender projection that transmits signals away from the cell body.
- **Myelin Sheath (e)**: This is a fatty layer that insulates the axon and speeds up signal transmission.
- **Node of Ranvier (f):** These are gaps between segments of the myelin sheath.

131)

NCERT-XI, Pg. # 232

132)

NCERT-XI, Pg. # 231

133)

NCERT-XI, Pg. # 233

134)

NCERT-XI, Pg. # 231

135)

NCERT-XI, Pg. # 232

136)

NCERT-XI, Pg. # 232

137)

The correct answer is 4. Nervous tissue.
Here's why:
- Ectoderm is the outermost of the three primary germ layers in the early embryo. It gives rise to various tissues and organs, including:
- Nervous tissue, Epidermis, Sensory organs

138)

Nurture Module No. 6 it is duct present in mid brain.

139)

NCERT-XI, Pg. # 232

140)

NCERT-XI, Pg. # 236

141)

Nurture Module No.6, related to lack of dopamine.

142)

NCERT-XI, Pg, # 236

143)

NCERT-XI, Pg, # 232

144)

NCERT-XI, Pg, # 232

145)

Nurture Module No. 6 Ventricle of brain are lined by ependymal epithelium

146)

Nurture Module No. 6, Regarding lose of cerebrum, specifically asked about temporal lobe of brain.

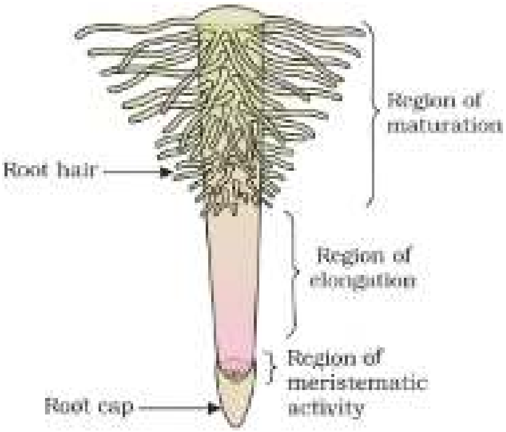
147)

Nurture Module No. 6, this is in reference of neuroglial cells.

148) NCERT, Pg # 66

149) NCERT-XI, Pg # 62

150) NCERT, Pg. # 59
Explain Question : Question asking about regions of root
Concept : This question is based on Root.
Solution : Given diagram represent the regions of the root-tip and labelled part A,B and C are -A-Region of maturation ,B-Region of elongation, C-Region of Meristematic growth



Final Answer : option (2)

151) NCERT, Pg. # 61
Question Explanation :
The question gives you three figures showing types of leaf arrangements (phyllotaxy).

Concept:
This question is based on Leaf: Phyllotaxy
Solution:
The correct answer is 2. Alternate, Opposite, Whorled
Explanation :
(i) Alternate - One leaf per node, alternating sides
(ii) Opposite - Two leaves per node, placed opposite each other
(iii) Whorled - More than two leaves from a single node
Final answer: 2. Alternate, Opposite, Whorled

152) NCERT Page No. 58

153) NCERT-XI, Pg. # 58

154) NCERT, Pg. # 62

155) NCERT-XI, Pg. 62

156) NCERT, Pg. 66
Explain Question:

Question asking about examples of drupe fruit
Concept:
This question is based on Fruit.
Solution:
In mango and coconut, the fruit is known as a drupe.
Final answer : 3

157) NCERT Pg. # 67

158) NCERT, Pg. # 61
Question Explanation :
The question asks about the growth behavior of the main axis in a racemose inflorescence, which is a common type of flower arrangement.
Concept:
This question is based on Inflorescence.
Solution:
The correct answer is 3 . Has unlimited growth
Explanation:
In racemose inflorescence:
• The main axis continues to grow (shows unlimited or indeterminate growth).
• Flowers are borne laterally in acropetal succession (older flowers at the base, younger at the top). **Final answer: 3. Has unlimited growth**

159)

NCERT, Pg. # 69
Explain Question : Question asking about floral characters of the Solanaceae family of angiosperms.
Concept : This question is based on Solanaceae family
Solution :
(a) In Solanaceae family, flowers are zygomorphic- **Incorrect**
In Solanaceae family, flowers are actinomorphic
(b) In Solanaceae family, ten stamens are present in a flower- **Incorrect**
In Solanaceae family, five stamens are present in a flower
(c) In Solanaceae family, seeds are endospermic.-**correct**
(d) In Solanaceae family, axile placentation is present.-**correct**
Final Answer : option (4)

160)

NCERT Page # 68

161)

NCERT, Pg. # 68
Explain Question : Question asking about selecting the correct floral formula according to the given floral diagram.(Solanaceae family.)
Concept : This question is based on Solanaceae family.

Solution : Given floral diagrams represent the family Solanaceae and correct floral formula is-

$\oplus ⚥ \; K_{(5)} \; \overbrace{C_{(5)} \; A_5} \; G_{(2)}$

Final Answer : option (3)

162)

NCERT, Pg. # 64

Explain Question : Question asking about floral characters of Malvaceae family

Concept : This question is based on Malvaceae family

Solution :

When filaments of stamens are united and anthers are free, the condition is called as........and found in family.... **Monoadelphous ,Malvaceae family.**

The presence of Monoadelphous stamens is a characteristic feature of the Malvaceae family.

Final Answer : option (4)

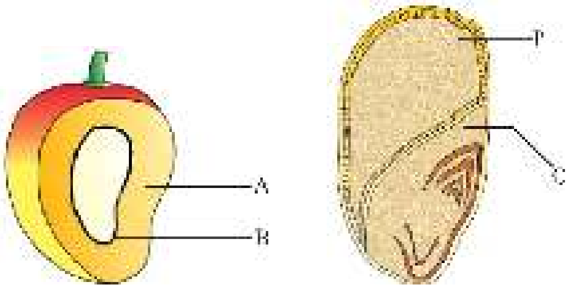
163) NCERT-XI, Pg. # 67

164) NCERT, Pg. # 66, 67

Question Explanation :

The question shows a diagram that includes two fruit wall layers (A and B) and two seed parts (P and Q)

You're being asked to identify what these labeled parts are.



Concept : This question is based on Fruits.

Solution :

A - Mesocarp : Middle fleshy layer of fruit wall.

B- Endocarp : Innermost fruit layer (often hard)

P- Endosperm : Food-storing tissue in seed

Q- Scutellum : Single cotyledon in monocot seed (e.g., maize)

Final Answer : (2)

165)

NCERT, Pg. # 107

Explain Question : The question asks: How many fatty acid chains are present in a molecule

of phospholipid ?
Concept : This question is based on Lipids
Solution : A phospholipid contains two fatty acid chains attached to glycerol and one phosphate group replacing the third fatty acid position.
Final Answer : option (1) Two

166)

NCERT, Pg. # 112
Solution:
Each **globin polypeptide chain** in hemoglobin folds into a **specific 3D shape**, forming a **tertiary structure**
The Correct answer is : 3. tertiary structure

167) NCERT, Pg. # 110
The correct answer is 4. Wax – Conjugated lipid.
Explanation: Chitin is a homopolysaccharide, Hyaluronic Acid is a heteropolysaccharide and Carotenoid is a chromolipid. However,Wax is not a conjugated lipid; it is a simple lipid. Conjugated lipids, like lipoproteins, consist of lipids bound to other molecules such as proteins or carbohydrates, which is not the case with wax. Therefore, Wax – Conjugated lipid is the odd one out.

168)

NCERT, Pg. # 111
Solution:
"A nucleotide has three components – a nitrogenous base, a pentose sugar and a phosphate group. A nucleoside is formed of a nitrogenous base and a sugar."

Answer:2 – Phosphate

169)

NCERT, Pg. # 112
Explanation:

A. **Assertion (A) is true** – The tertiary structure of a protein gives it a 3D conformation, which is crucial for its function.

B. **Reason (R) is false** – Hydrogen bonds and disulfide bonds help in stabilizing a single protein molecule, not in holding two or more protein molecules together. The correct interactions that hold multiple protein molecules together are non-covalent interactions in the quaternary structure

   Correct Answer: 1 ((A) is true, but (R) is false.)

170) NCERT, Pg. # 106
Explanation:
• Arachidonic acid ($C_{20}H_{32}O_2$) is a polyunsaturated fatty acid (PUFA) with 20 carbon atoms and four double bonds
Correct Answer: 2 (Arachidonic acid)

171) NCERT, Pg. # 109
Correct Answer: 1. (Both (A) & (R) are true and (R) is the correct explanation of (A).)
Explanation:
• Assertion (A) is true:
Enzymes, being proteins, lose their activity at high temperatures because their three-dimensional structure is disrupted.
• Reason (R) is true:
Proteins denature when exposed to high temperatures, meaning they lose their specific folding and hence their functional conformation.
• Relationship:
The denaturation of proteins directly leads to the loss of enzymatic activity, making (R) the correct explanation for (A)

172)

NCERT, Pg. # 107
The correct answer is: **3. Phospholipid**

Explanation:

A. **Lecithin** is a common name for **phosphatidylcholine**, which is a type of **phospholipid**.

B. Phospholipids are major components of cell membranes, consisting of:

    A. Glycerol backbone

    B. Two fatty acid chains

A phosphate group linked to a nitrogenous compound (like choline in lecithin)

173)

NCERT, Pg. # 107
Let's identify the given amino acid structure:
The presence of the $-CH_2-OH$ side chain is characteristic of the amino acid serine.
Non-essential amino acids are those that can be synthesized by the human body . Serine is a non-essential amino acid.
Correct option = 4

174)

NCERT, Pg. # 108
Lectins are carbohydrate-binding proteins. c.
Concanavalin A is a well-known example of a lectin.

Essential oils are concentrated hydrophobic liquids containing volatile aroma compounds from plants.
b. Lemon grass is famous for its essential oil, which has a distinct citrusy aroma.
Pigments are substances that produce color.
d. Anthocyanin is a type of pigment responsible for many of the red, purple, and blue colors we see in fruits, vegetables, and flowers.
Alkaloids are naturally occurring chemical compounds containing basic nitrogen atoms.
a. Codeine is an alkaloid derived from the opium poppy and is used as a pain reliever and cough suppressant.
Correct option = 2

175) NCERT Pg # 110

176) NCERT, Pg. # 112
The correct option is: **(3) a–Secondary structure, b–Tertiary structure**
Explanation:
Only in **the secondary structure,** there is a complex folding of the linear polypeptide chain int a specific structure which is the Alpha helical structure. When it is repeated, they form a pleated sheet-like structure. This forms the beta pleated sheet. This has hydrogen bonds in addition to the peptide bonds that provide stability to the protein and this is the secondary structure of the protein.
When there is more folding of the secondary proteins, they form **the tertiary structure.** This is the functional structure of proteins. Tertiary structure gives us a 3-dimensional view of proteins. Proteins of tertiary structure are highly folded to give a globular appearance.
Hence, option (3) is correct.

177) NCERT, Pg. # 104

178) NCERT, Pg. # 105

179)

NCERT, Pg. # 109, 110

180)

NCERT, Pg.# 104